

# Audit Logging and Compliance

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## Meeting Regulatory and Security Requirements

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Audit logging is essential for compliance (SOC2, SOX, HIPAA, PCI-DSS) and security operations. This notebook covers querying audit logs, building compliance reports, and conducting access reviews.

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# Prerequisites

| Requirement           | Details                                                |
|-----------------------|--------------------------------------------------------|
| Dynatrace Environment | SaaS with Gen3 IAM and audit logging enabled           |
| Permissions           | <code>logs:read</code> with access to audit log bucket |
| Data Retention        | Audit logs retained per compliance requirements        |

## 1. Audit Log Fundamentals

Dynatrace audit logs capture all security-relevant events in your environment.

### What Gets Logged

| Category         | Events                                  |
|------------------|-----------------------------------------|
| Authentication   | Logins, logouts, MFA events, SSO        |
| Authorization    | Access attempts, permission checks      |
| User Management  | User creation, modification, deletion   |
| Group Management | Group changes, membership updates       |
| Policy Changes   | Policy creation, modification, deletion |
| Token Operations | Token creation, revocation, usage       |
| Configuration    | Settings changes, API calls             |

## Audit Log Fields

| Field            | Description   | Example                  |
|------------------|---------------|--------------------------|
| timestamp        | Event time    | 2026-01-26T14:30:00Z     |
| log.source       | Log category  | audit                    |
| content          | Event details | JSON with action details |
| dt.security.user | Acting user   | user@company.com         |
| dt.source_entity | Source entity | Client IP, service       |

## Retention Requirements

| Compliance | Minimum Retention         |
|------------|---------------------------|
| SOC2       | 1 year                    |
| SOX        | 7 years                   |
| HIPAA      | 6 years                   |
| PCI-DSS    | 1 year online, 1+ archive |
| GDPR       | As long as necessary      |

## 2. Authentication Event Monitoring

Track who is accessing your Dynatrace environment and how.

```
// All authentication events in the last 24 hours
fetch logs, from: now() - 24h
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login") or matchesPhrase(content,
"authentication")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Failed login attempts – critical for security monitoring
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "failed") and matchesPhrase(content, "login")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Login activity by hour to identify patterns
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login")
| summarize logins = count(), by: { hour = bin(timestamp, 1h) }
| sort hour asc
```

Authentication Patterns to Monitor

| Pattern                 | Indicates             | Query Focus         |
|-------------------------|-----------------------|---------------------|
| Multiple failed logins  | Brute force attempt   | Failed + same user  |
| Off-hours logins        | Unusual access        | Timestamp analysis  |
| New location logins     | Account compromise    | IP/location change  |
| Rapid successive logins | Automation or sharing | Time between logins |
| Login after offboarding | Process failure       | User status check   |

```
// Off-hours login detection (outside 8am-6pm)
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login")
| fields timestamp, content
| filter hour(timestamp) < 8 or hour(timestamp) > 18
| sort timestamp desc
| limit 50
```

3. Authorization and Access Events

Track permission checks and access attempts to sensitive resources.

```
// Access denied events – indicates permission issues or attacks
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "denied") or matchesPhrase(content,
"forbidden") or matchesPhrase(content, "unauthorized")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Permission changes and grants
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "permission") or matchesPhrase(content,
"grant") or matchesPhrase(content, "role")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

## Access Event Categories

| Event Type               | Security Relevance                   | Monitoring Priority |
|--------------------------|--------------------------------------|---------------------|
| Access denied            | Permission misconfiguration or probe | High                |
| Privilege escalation     | Critical security event              | Critical            |
| Cross-environment access | Boundary violation                   | High                |
| Admin action             | Privileged operation                 | Medium              |
| Bulk data access         | Exfiltration risk                    | High                |

## 4. Configuration Change Tracking

Monitor all changes to IAM configuration.

```
// User creation and deletion events
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "user") and (matchesPhrase(content,
"created") or matchesPhrase(content, "deleted"))
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Group membership changes
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "group") and (matchesPhrase(content, "added")
or matchesPhrase(content, "removed") or matchesPhrase(content, "membership"))
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Policy changes - critical for compliance
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "policy") and (matchesPhrase(content,
"created") or matchesPhrase(content, "modified") or matchesPhrase(content,
"deleted"))
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Token creation and revocation
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token") and (matchesPhrase(content,
"created") or matchesPhrase(content, "revoked"))
| fields timestamp, content
| sort timestamp desc
| limit 100
```

## Change Tracking Summary

| Change Type      | Audit Requirement | Review Frequency |
|------------------|-------------------|------------------|
| User lifecycle   | Log all           | Weekly           |
| Group changes    | Log all           | Weekly           |
| Policy changes   | Log + approve     | On change        |
| Admin actions    | Log + review      | Daily            |
| Token operations | Log all           | Monthly          |

## 5. Compliance Reporting Patterns

Standard queries for compliance frameworks.

### SOC2 - Access Control Evidence

SOC2 requires evidence of access controls and monitoring.

```
// SOC2: User access review – all users who accessed system in period
fetch logs, from: now() - 90d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login")
| summarize last_login = max(timestamp), login_count = count(), by: { content
}
| sort last_login desc
| limit 500
```

```
// SOC2: Administrative action log
fetch logs, from: now() - 90d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "admin") or matchesPhrase(content, "created")
or matchesPhrase(content, "deleted") or matchesPhrase(content, "modified")
| fields timestamp, content
| sort timestamp desc
| limit 500
```

## SOX - Segregation of Duties

SOX requires segregation of duties and change authorization.

```
// SOX: Privileged user activity (admin actions)
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "admin")
| summarize action_count = count(), by: { content }
| sort action_count desc
| limit 100
```

```
// SOX: Configuration changes requiring approval
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "policy") or matchesPhrase(content,
"settings")
| filter matchesPhrase(content, "modified") or matchesPhrase(content,
"created") or matchesPhrase(content, "deleted")
| fields timestamp, content
| sort timestamp desc
| limit 200
```

## HIPAA - PHI Access Logging

HIPAA requires logging of access to protected health information.

```
// HIPAA: All data access events for PHI-related resources
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "read") or matchesPhrase(content, "access")
or matchesPhrase(content, "query")
| fields timestamp, content
| sort timestamp desc
| limit 500
```

```
// HIPAA: Access anomalies - bulk data access detection
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| summarize event_count = count(), by: { hour = bin(timestamp, 1h), content }
| filter event_count > 100
| sort event_count desc
| limit 50
```



# PCI-DSS - Cardholder Data Access

PCI-DSS requires tracking access to cardholder data.

```
// PCI-DSS: Failed access attempts (potential attack indicator)
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "denied") or matchesPhrase(content, "failed")
| summarize failure_count = count(), by: { day = bin(timestamp, 1d) }
| sort day desc
```

## Compliance Query Reference

| Framework | Key Reports                                | Frequency    |
|-----------|--------------------------------------------|--------------|
| SOC2      | User access, admin actions, logical access | Quarterly    |
| SOX       | Privileged users, change authorization     | Quarterly    |
| HIPAA     | PHI access, anomaly detection              | Monthly      |
| PCI-DSS   | Failed access, CDE access                  | Daily/Weekly |
| GDPR      | Data access, consent tracking              | On request   |

## 6. Access Review Workflows

Periodic access reviews are required by most compliance frameworks.

### Access Review Process

| Step | Action                      | Owner     |
|------|-----------------------------|-----------|
| 1    | Generate user access report | IAM Admin |
| 2    | Distribute to managers      | IAM Admin |
| 3    | Review and certify access   | Managers  |
| 4    | Remove inappropriate access | IAM Admin |
| 5    | Document review completion  | IAM Admin |

```
// Access review: Users who haven't logged in for 90 days
fetch logs, from: now() - 180d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login")
| summarize last_login = max(timestamp), by: { content }
| filter last_login < now() - 90d
| sort last_login asc
| limit 100
```

```
// Access review: High-activity users (potential over-privileged)
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| summarize action_count = count(), by: { content }
| sort action_count desc
| limit 50
```

## Access Review Checklist

- ☐ All users verified as current employees
- ☐ Access levels appropriate for job function
- ☐ No separation of duties violations
- ☐ Privileged access justified and approved
- ☐ Inactive accounts identified for removal
- ☐ Service accounts still required
- ☐ Review documented and signed off

## Review Frequency by Access Level

| Access Level  | Review Frequency | Approver            |
|---------------|------------------|---------------------|
| Viewer        | Annually         | Manager             |
| Editor        | Semi-annually    | Manager + Security  |
| Admin         | Quarterly        | Director + Security |
| Account Admin | Monthly          | CISO                |

# 7. Building Compliance Dashboards

Create dashboards for ongoing compliance monitoring.

## Dashboard Components

| Section          | Metrics                   | Chart Type |
|------------------|---------------------------|------------|
| Authentication   | Logins/failures over time | Line chart |
| Access Denied    | Failed access attempts    | Bar chart  |
| User Changes     | Create/delete/modify      | Table      |
| Admin Activity   | Privileged actions        | Table      |
| Token Operations | Creates/revocations       | Table      |

```
// Dashboard: Authentication trend (for line chart)
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "login")
| summarize logins = count(), by: { day = bin(timestamp, 1d) }
| sort day asc
```

```
// Dashboard: Failed access by day (for bar chart)
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "denied") or matchesPhrase(content, "failed")
| summarize failures = count(), by: { day = bin(timestamp, 1d) }
| sort day asc
```

```
// Dashboard: Recent administrative actions (for table)
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "created") or matchesPhrase(content, "deleted") or matchesPhrase(content, "modified")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

# Alert Thresholds

Configure alerts based on audit log patterns:

| Condition        | Threshold                  | Severity |
|------------------|----------------------------|----------|
| Failed logins    | > 5 in 10 min same user    | Warning  |
| Failed logins    | > 20 in 10 min any         | Critical |
| Admin action     | Any outside business hours | Warning  |
| Policy change    | Any                        | Info     |
| User deletion    | Any                        | Info     |
| Bulk data access | > 1000 queries/hour        | Warning  |

# Next Steps

With compliance monitoring in place, proceed to multi-environment management:

## Recommended Path

1. **IAMADM-08: Multi-Environment IAM** - Scale IAM across environments
2. **IAMADM-09: Troubleshooting Access Issues** - Debug permission problems

## Compliance Checklist

Before moving on, ensure you have:

- ☐ Identified applicable compliance frameworks
- ☐ Created queries for required audit reports
- ☐ Established access review schedule
- ☐ Built compliance monitoring dashboard
- ☐ Configured alerts for critical events
- ☐ Documented audit log retention policy

# Summary

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In this notebook, you learned:

- Audit log fundamentals and retention requirements
  - Authentication event monitoring patterns
  - Authorization and access event tracking
  - Configuration change monitoring
  - Compliance-specific query patterns (SOC2, SOX, HIPAA, PCI-DSS)
  - Access review workflows
  - Compliance dashboard components
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# References

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- [Audit Logs](#)
  - [DQL Reference](#)
  - [SOC2 Trust Principles](#)
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